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SUPERSEDING
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27 November 1984

MILITARY SPECIFICATION

LATHE, ENGINE, FLOOR MOUNTED, 17-INCH SWING

This specification is approved for use by the Armament Research, Development and Engineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid bed, rigid base, floor mounted, metal cutting, electric-motor driven 17-inch minimum swing, engine lathe with standard equipment and accessories as described herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-L-800	-	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood

MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-C-10924	-	Grease, Automotive and Artillery

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-16173 - Corrosion Preventive Compound
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-681 - Identification Coding and Application of Hookup and Lead Wire
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing, and Marking

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the specification.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B5.1 - T-Slots, Their Bolts, Nuts, and Tongues
B5.8 - Chucks and Chuck Jaw
B5.9 - Spindle Noses
B5.10 - Machine Tapers

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B5.16 - Accuracy of Engine and Tool Room Lathes
B11.6 - Lathes-Safety Requirements for the Construction, Care, and Use of Lathes

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360.02 - Machine Tool Gearing
AGMA 390.03 - Gear Handbook, Volume I, Gear Classification, Materiels, and Measuring Methods for Unassembled Gears
AGMA 600.01 - Metric Usage

(Application for copies should be addressed to American Gear Manufacturers Association, One Thomas Circle, Washington, DC 20005.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

(Applications for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E380 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

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3.2 Design. The lathe shall be new and one of the manufacturer's current model(s) conforming to the accuracy requirements for an engine lathe as specified herein. The lathe shall be capable of performing turning, boring, facing, and threading operations within the capacities of the size specified. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. The lathe shall be designed and constructed to provide convenient and safe operation for the machine operator. The lathe design shall incorporate all the requirements specified herein to the best commercial practices of the domestic manufacturing industry. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, repair, or replacement without the use of special tools and fixtures. The lather bed shall be designed to prevent chips and shavings from collecting inside of any cavities or crevices. Modifications to manufacturer's standard designs to achieve requirements specified herein are not permissible, if they result in deviations from good design practice. Examples of such impermissible modifications are placing a 15 Hp motor in a power train designed and normally used for 3 Hp, or raising the headstock on a 13 inch lathe to achieve a 17 inch swing.

3.2.1 Design, manufacture, and construction measurement systems. The manufacturer may use either the US Customary System of Units (US) or the International System of Units (SI) in the design, manufacture, and construction of the lathe. In this specification, all measurements, dimensions, sizes, and capabilities are given in the US system. These measurements may be converted to the SI system through the use of the conversion factors and methods specified in ASTM E380. Regardless of which system is used by the manufacturer, all measurements, dimensions, sizes, and capabilities shall be specified in the same system, US or SI, but not both. The manufacturer is encouraged to provide measuring and indicating devices such as scales, carriage stops, dial indicators, pressure gauges, and all other similar devices in the US inch and the metric (SI) system of measurement. When a single system is used, the US system is acceptable but the SI system is preferred.

3.2.2 Hardness integrity. Gears, machine components, and ancillary equipment shall be hardened in accordance with the requirements herein. The machine part or component shall conform to the necessary size, configuration, and finish as dictated by the mechanical application. Gears, machine components, or ancillary equipment shall not be hard plated to achieve the specified size or hardness levels.

3.2.3 Gears. All gear and pinion designs selected for use in the lathe or its components shall be manufactured in accordance with AGMA 390.03, AGMA 360.02, and AGMA 600.01. All back gears, gears in the headstock, gears in the quick change gear box, and gears in the drive train shall be hardened steel. The required back lash between mating teeth shall be determined in accordance with AGMA 390.03, Part I, Section 8. The hardness of the gears shall be not less than Rockwell C-48. The gears shall be of the proper width throughout the speed ranges without failure for the extended life of the machine.

3.2.4 Threads. The design, manufacture, and configuration of threads shall be in conformance with FED-STD-H28 and the applicable detailed standards referenced therein. No testing in FED-STD-H28 shall be required.

3.2.5 Lubricating systems. All bearings (except sealed for life, permanently lubricated type), matching gears, and all other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be

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so located as to be readily accessible. The supplier shall include and install all lubricants, fluids, greases, and other compounds in the lathe, in accordance with the best commercial practices of the lathe industry. Vent openings in the lubrication system shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM," shall be attached to all temporary caps or plugs intended to be vented to the atmosphere. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, the appropriate lubricant shall be provided in a sealed container and shipped with each lathe. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to install the furnished lubricant before operation of the equipment. The operator's manual shall also reflect the above warning and procedure.

3.2.6 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with the best commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components of the overall assembly. The use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permissible. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2.7 Operating controls. All operating controls for the lathe shall be directly mounted on the lathe in a location convenient to the machine operator's normal working position.

3.3 Construction. The lathe shall be constructed of parts and materials which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without distortion. All parts and mechanisms subject to periodic adjustment, repair, or replacement shall be accessible, without disturbing, modifying or damaging the lathe or its component parts. Construction of the lathe shall be complete so that when installed and connected to a power supply, it will be ready for operation. All screws, pins, or similar parts shall be installed to prevent loss of tightness.

3.3.1 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances. When industry or Government standards are applicable, all replaceable parts shall conform to the standards. Parts which have a mortality rate, during normal operation, shall be identified by the supplier's part number and if purchased by the supplier, the Original Equipment Manufacturer's (OEM) part number. All replaceable parts of the lathe shall be interchangeable with an item of the same part number without requiring modification of the part or its mating components. The part numbers shall be included in the repair parts manual, to be used for replacement parts requirements. These replacement parts shall provide the same or better form, fit, function, and performance as the part initially installed in the lathe.

3.3.2 Safety provisions. Covers, guards, or other safety devices shall be provided with the lathe and the devices shall not interfere with the operation of the lathe. As a minimum. point of operation guarding for the lathe shall include a chuck guard, a splash guard, and a travel guard. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components,

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mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of 29CFR, Chapter XVII, Part 1910 that are applicable to the lathe itself. In addition, the machine shall comply with all requirements of ANSI B11.6 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply.

3.3.3 Protective finish. The protective finish shall be paint. All surfaces to be painted shall, immediately prior to painting, be cleaned, free of all foreign matter, and dried. The protective paint coating shall be at least 2.0 mils (0.002 inch) thick. The paint adhesion shall be tested by making two intersecting cuts in the paint which penetrate through to the base metal. The cuts shall be not less than 2 inches in length. Fiberglass tape shall be applied over the two intersecting cuts and then shall be removed without any peeling of the paint. The surface hardness of the paint shall be between the industrial standards of H to HB lead. Where no natural finish of the material or the finish obtained from heat treatment is permissible provided the surfaces are free from scale or corrosion.

3.3.4 Ways. The ways shall be adequately protected to prevent chips(s), dirt, coolant, and other hazards from causing damage or resulting in inaccurate functioning of any interfacing surface members of the lathe. The machine ways shall be hardened and ground in accordance with the requirements specified herein. The finish on the machine ways shall be not more than 32 microinches, arithmetic average (aa).

3.3.5 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, or filling with solder or paste shall be used for reclaiming any defective parts.

3.3.6 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.7 Fastening devices. All screws, pins, bolts, and similar fasteners which affect machine operation or prevent loss of fluids shall be installed with means of preventing loss of tightness. Those fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.8 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the best commercial practice of the lathe manufacturing industry.

3.4 Dimensions and characteristics. The dimensions and characteristics shown in table I are the established Government requirements and shall be met or exceeded for the lathe specified herein. Requirements are not less than figures shown. Where a range is shown, the requirement is from the smaller figure or less to the larger figure or more.

3.4.1 Machine tolerances. The lathe shall conform to the accuracy requirements as specified, explained, and diagramed in ANSI B5.16. Machine tolerance shall be tested in accordance with ANSI B5.16.

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TABLE I. Sizes and Capacities

Machine characteristics	Dimensions in inches except as noted
Capacities:	
Swing over bed	17
Swing over cross slide	9
Center distance	54
Main drive motor Hp, continuous duty	7 1/2
Headstock	
Spindle type and size	D1-8 or L2
Diameter of thru-hole	1 9/16
Center size, morse taper	No. 5
Number of speeds	16
Speed range, RPM	40 to 1450
Carriage	
Cross slide travel	9 5/8
Compound rest travel	3
Combination inch and metric gearbox:	
Thread selections, English	48
Thread selections, metric	48
Feed range, IPR	0.0025 to 0.052
Thread range, per inch	2 to 72
Lead in mm	0.40 to 6
Tailstock:	
Spindle diameter	2 1/4
Spindle travel	5
Center size, morse taper	No. 4
Tailstock set over, each side of center line	1/2
Accessory Equipment:	
Taper attachment:	
Taper per foot	3 1/2
Length turned at one setting	15
Face Plate diameter, large	17
Face Plate diameter, small	9
Tool post, to accommodate tool holder size	5/8 by 1 3/8
Steady rest capacity, diameter	1/2 to 5
Follow rest capacity, diameter	1/2 to 2 1/2
Chuck diameter (inches), 3-jaw	12
Chuck diameter (inches), 4-jaw	15

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3.5 Components. The lathe shall consist basically of the following components.

3.5.1 Bed and base. The lathe shall have a bed and base structure that supports the headstock, tailstock, carriage, and other related equipment. The bed shall be of the solid-bed type. The bed and base shall be a ferrous casting with integral cross ribs, having sufficient strength and rigidity to maintain all the lathe components in alignment in accordance with ANSI B5.16 alignment requirements, while turning parts. The turned parts shall be accurate for rough and finish cutting as specified in 3.9.1 and 3.9.2. The bed shall have V-ways, or a combination of "V" and flat ways to provide a sliding surface for the longitudinal movement of the carriage and assure alignment of the headstock and tailstock. The ways for the carriage longitudinal movement shall have a surface hardness of not less than Rockwell C-42 and shall be ground to minimize wear and maintain accuracy of alignment. The ways may be either cast integrally with the bed structure or may be replaceable way inserts, or a combination thereof. Means shall be provided to lubricate the ways. The ways shall have covers or wipers that effectively prevent entrance of coolant, chips, or other extraneous materials between the ways and carriage slide. The bed shall have a sufficient number of covered access openings to facilitate inspection adjustment, and repair of components located within the structure. Means of leveling the machine shall be provided.

3.5.2 Headstock. The headstock shall be affixed to the lathe bed by means of keys or pins and bolts. An adjustable headstock shall not be acceptable. The spindle and spindle driving gear shafts shall be supported by tapered roller bearings (see 3.5.2.1). The spindle shall be operable in both forward and reverse directions of rotation. The headstock shall rigidly support the spindle and gear shafts in a secure manner to prevent deflection that would affect accuracy under operating and full power cutting loads. Spindle drive gears shall have a surface hardness of not less than Rockwell C-48. Easy access to the drive mechanism shall be provided. During setup, the spindle shall be free to rotate manually, and there shall be provisions for retaining the spindle in any desired position to facilitate the mounting or operations of chucks, face plates and other spindle mounted attachments. Headstocks which require a spindle locking device shall have an interlock safety mechanism to prevent power application while the spindle is locked. Headstocks, if belt driven, shall have means for changing, adjusting, and maintaining proper belt tension without loss of spindle alignment or adjustment. The headstock shall be provided with a self-contained lubrication system. The headstock lubrication system shall provide positive means of delivering oil to bearings, gears, and sliding or rotating parts that require lubrication. Splash systems, if used, shall have a provision for ensuring adequate lubrication of all bearings and parts requiring lubrication. Gearing for the transmission shall run in a bath of oil with provisions for checking and maintaining the oil level in the transmission. An identification plate shall indicate shift level positions for designated spindle speeds. The indicating plate shall be readily visible from the operator's normal working position. Headstock speeds shall be manually adjustable by either a shift lever, with selected speeds, or by an infinitely variable speed selector switch. Should the shift lever speed change system be furnished, no less than 16 selected speeds shall be provided, as specified in table I. Speeds shall be an approximate geometric progression. The infinitely variable speed headstock shall provide spindle speed selections for all speeds within the range of 40 to 1450 Revolutions Per Minute (RPM). Speed selections shall be possible while cutting to obtain the optimum speed/feed ratio under various cutting conditions. The variable speed headstock shall be designed and constructed to eliminate the need to run the lathe through its speed range to prevent sticking of the spindle. The headstock shall meet or exceed the accuracy requirements of ANSI B5.16.

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3.5.2.1 Headstock spindle. The headstock spindle shall be designed for using draw-in collets, chucks, and face plates which in turn support and provide rotation of the workpiece. The spindle shall be supported by tapered roller bearings, located in the front and rear positions of the spindles. (Spindle designs incorporating at least three spindle bearings are acceptable.) Thrust bearings shall have means for adjustment. The spindle shall have a hole through the center to accommodate extra long work stock. The spindle nose shall be the D1-8 Camlock type as specified in table I and shall conform to ANSI B5.9. The spindle shall have a taper conforming to ANSI B5.10. Taper size and hole diameter shall be as specified in table I. The spindle nose shall provide locating surfaces for lathe center, face plates, and chucks. The inner and outer surfaces of the spindle nose on which work holding devices are mounted shall be made of steel and shall have a surface hardness of not less than Rockwell C-50. The headstock spindle shall meet the accuracy requirements of ANSI B5.16.

3.5.2.2 Spindle drive. The spindle shall be driven through either an all gear of combination gear and drive belt drive system by an electric motor as specified in table I. The combination drive shall provide nonslip positive belt drive in the high speed range. The intermediate and low speeds shall be gear driven. The spindle speeds shall be in approximate geometric progression. The spindle speed shall operate in both directions of rotation.

3.5.3 Power feed. Power feeds and thread leads shall be provided through a totally enclosed, quick change, English and metric, gear box. The feed screws shall be fitted with micrometer dials graduated in increments of not more than 0.001 inch, or 0.02 millimeter, or both. The feed screw micrometer dials shall be accurate indicators of tool movement and stock removal. The dials shall be permanently and legibly etched or engraved on a nonglare background. The dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position. The power feed system shall contain a feed rod, a hardened and ground lead screw, and a geared feed mechanism. The lead screw shall be hardened to not less than Rockwell C-38 and shall meet the alignment requirements of ANSI B5.16. The longitudinal feed shall be by rack and pinion, with the rack mounted on the underside of the front ways, meshing with the pinions of the apron. The rack pinion tool hardness difference, or use of different metals shall be designed to impart most wear to the component most easily replaced. The longitudinal and cross feeds shall be in both directions of travel with right and left hand threading capability. A separate feed rod and leadscrew shall be used and the feed motion shall be accomplished without use of the threads to retain threading accuracy. Automatic lubrication shall be provided for the quick change gearing. A shear pin shall be provided to protect the feed mechanism from damage by accidental overload. An interlock shall prevent simultaneous engagement of the thread and feed controls. A feed and thread chart shall be written in the English language and shall be readily visible to the operator's normal working position to show control settings. The half nut shall be made from material which is somewhat softer than the leadscrew, like brass or bronze that will allow the half nut to wear more quickly than the leadscrew.

3.5.4 Carriage assembly. The carriage assembly shall include the apron, tool post, cross slide, saddle, and compound rest. The carriage, cross slide, and compound rest shall be in accordance with ANSI B5.16 and accurately fitted to the mating way surfaces and shall be providing smooth, constant feed motion under power and shall have adjustable gibs to permit compensation for wear. A carriage clamping device shall securely lock the carriage in any position along the bed when cross feeding. A carriage reversing mechanism shall allow feed movement in both

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directions without changing direction of the spindle rotation. The reversing control shall be easily operated and positive acting, located within easy reach from the operator's normal work position. Way wipers shall be provided on the carriage wings to prevent chips and abrasives from entering between the carriage and ways. The carriage shall be provided with the necessary means for attaching accessories such as follow rests and a taper attachment.

3.5.6 Cross feed screws. Threads of the cross slide and top slide feed screws shall have a surface hardness of not less than Rockwell C-45. The cross slide and the top slide feed screw threads shall be Acme of square form (see FED-STD-H28) and shall be protected from chips and other foreign matter by the construction of the cross slide or by a suitable guard. The cross slide feed screw shall be supported by anti-friction thrust bearings and shall have an adjustment for backlash.

3.5.7 Apron. The apron shall contain the feed mechanism, threading mechanism, operating controls, and automatic lubrication system for the apron and carriage components. The apron shall be double wall construction providing inboard and outboard supports for gear shafts and studs. Gearing in the apron shall provide for both the longitudinal and cross feeds in the manual and power operational modes. Feeds shall be engaged by means of a clutch, operated from the front of the apron. The clutch shall be a safety type to protect the feed mechanism against damage. A selector level shall be provided on the front of the apron for engaging the cross and longitudinal feeds. An interlock shall be incorporated in the apron to prevent engagement of the feed and half nut at the same time. Handwheels shall automatically disengage to assure nonrotation while the carriage is under power. Rotating shafts shall be supported at both ends by ball or roller bearings. The reversing control for the spindle shall be located on the apron.

3.5.7.1 Leadscrew reverse lever. The leadscrew reverse shall be provided in a convenient location within reach from the operator's normal work station to permit reversing the direction of the carriage movement without disengaging the leadscrew half nut, or reversing/stopping the spindle.

3.5.8 Tailstock. The lathe shall have a tailstock which is accurately fitted to the bedways of the lathe. The tailstock shall be easily moved along the bedways and shall have a quick action clamp for locking the tailstock in all locations along the bedways. The tailstock spindle shall meet the alignment and accuracy requirements of ANSI B5.16. The tailstock shall have alignment markings which show proper center position for straight turning, and for measurement of the distance, when turning tapers without the aid of a taper attachment. The amount of setover shall conform to the requirements of table I. The alignment marking shall have a zero point, shall extend each side of zero to the setover distance specified in table I, and shall be marked in increments not greater than 1/16 of an inch. A means shall be provided to lock the tailstock spindle at any point of spindle travel without deflection. Tailstock spindle travel shall be accomplished by a handwheel, without jamming when either fully extended or fully retracted. The total distance between the fully extended and fully retracted positions of the tailstock spindle shall meet the travel requirements of table I. The inside of the spindle shall be taper bored to hold the tailstock center, drill chuck, or other tapered, tang type cutting tools. These tang type tools shall eject when the spindle is retracted to the end of spindle travel. The taper shall conform to the applicable specifications of ANSI B5.10 and table I. The tailstock shall be hardened to a hardness of not less than Rockwell C-48 and shall be graduated in increments not greater than 1/16 inch for measuring spindle travel. The lathe design shall provide for lubrication of the surface between the tailstock and the bedways. The design shall also provide for lubrication between the tailstock spindle and the tailstock.

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3.6 Electrical system. The electrical system shall conform to the applicable requirements of the NFPA 70 and NFPA 79. A disconnect switch shall be provided to disengage all incoming power and shall be separate from the machine's on/off switch. The machine shall be capable of operating on 230/460 volt, three phase, 50/60 Hertz power and shall be initially wired to operate on 460 volts. An identified terminal shall be provided, suitable for connecting the proper size grounding conductor in accordance with the requirements for the specified power source.

3.6.1 Motor. The main drive shall be not less than 7 1/2 Hp with permanently sealed ball or roller bearings rated for continuous duty operation and shall meet the requirements for a drip-proof enclosure. The motors shall conform to the requirements of the NEMA MG-1.

3.6.2 Color coded electrical circuitry. The wiring in the electrical system shall have individual point to point color coded wires in accordance with the System II requirements of MIL-STD-681. The electrical system diagrams/schematics provided with each lathe shall be an exact duplication of the electrical system and wiring in the lathe. Diagrams/schematics shall clearly identify the individual, point to point color coded wire and wire termination points located in the electrical system.

3.6.3 Electrical power supply cable. An electrical power supply cable shall be installed on each machine. The electrical power supply cable shall be not less than 10 feet in length and shall terminate in not less than 1 inch of stripped tinned wire, without a receptacle plug. The electrical power supply cable shall include sheathing/covering in accordance with NFPA 70 and 29CFR, Chapter XVII, Part 1910. The electrical power supply cable shall have a voltage/potential and ampere/current carrying capacity exceeding that required by the electrical power consuming devices, including electrical motors, of the lathe in accordance with NFPA 70.

3.6.4 High voltage requirement. The entire electrical system of the lathe shall withstand, without damage, malfunction, breakdown, arcing, surface or air discharge, at least 2500 delivered volts root mean square (rms), 60 Hertz of electrical energy, at ambient temperature. The electrical energy shall be applied instantaneously at full voltage of at least 2500 delivered volts, using at least a 500 volt ampere transformer and shall be maintained for at least 15 seconds.

3.6.4.1 Power input. The test voltage shall be applied to the power input terminals of the lathe with no protection for solid state components. The ground connection from the power source shall be the same ground connection as the lathe. No arcing or discharge is acceptable.

3.6.4.2 Internal circuits. Solid state control components may be damaged as a result of high voltage and is not cause for rejection. Damage to solid state control components may be avoided by wiring around or bypassing them terminal to terminal when subject to the requirements in 3.6.4. No arcing or discharge from solid state control components is acceptable.

3.6.5 Electromagnetic interference. When specified (see 6.2.1 and 6.6), the lathe shall comply with the electromagnetic interference control requirements of MIL-STD-461 for Class C3, Group II items.

3.7 Elapsed time meter. The lathe shall be fitted with a standard meter to measure the accumulative operating time of the main drive motor. The meter shall be calibrated to indicate in hours. The indicator shall be the nonresetting type and

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shall have a range of 1 hour to 9999 hours. The case containing the meter shall be designed to prevent entrance of dust and moisture. The meter shall be mounted to withstand normal vibration and shock generated by the machine.

3.8 Accuracy. The lathe shall be constructed to a rigidity that is sufficient to meet all the accuracy requirements of the test standards for 12 inch to 18 inch engine lathes in ANSI B5.16 after being subjected to the cutting power requirements in 3.9, 3.9.1, and 3 9.2.

3.9 Cutting power. The lathe shall develop the power required to perform rough and finish cutting operations under the conditions stated herein. The cutting operations shall be performed on SAE 1045, round, plain carbon steel bar, of not less than 2.00 inches in diameter, with a brinell hardness of not less than 160.

3.9.1 Rough cutting. The rough cut shall be no less than .080 inches deep and no less than 3.00 inches long. The cut shall be made at a feed rate of not less than 0.015 inches per revolution at a cutting speed of not less than 350 surface feet per minute.

NOTE: To convert surface feet per minute (V) to lathe revolutions per minute (N), divide V by .262 times the diameter to be cut, in inches; i.e.,

$$N = \frac{V}{DIA \times .262}$$

The rough turned diameter shall show no evidence of chatter and shall meet a total tolerance requirement of 0.005 inches or less for both out-of-round and taper per foot.

3.9.2 Finish cutting. Finish cutting operations shall be performed on the same SAE 1045 steel bar as the rough cut. The finish cut shall be no less than .020 inches deep and no less than 3.00 inches long. The finish cut shall be made at a feed rate of not less than 0.005 inches per revolution at a cutting speed of not less than 400 surface feet per minute. The finish turned diameter shall meet the requirements specified in ANSI B5.16.

3.10 Performance. The lathe and its components operator's controls and safety devices shall function properly under the following conditions:

3.10.1 Turning. A bar of low carbon steel (SAE 1020) of not less than 2 inches in length using a cutting speed of not less than 300 surface feet per minute. The turned diameter shall conform to the accuracy of test 21 of ANSI B5.16.

3.10.2 Metal removal turning. A bar of low carbon steel (SAE 1020) of not less than 5 inches in diameter and 15 inches in length shall be held in the four-jaw chuck and supported at the other end by the tailstock center. A cut shall be made for a length of not less than 12 inches using a single point carbide tip turning tool and a cutting speed of not less than 267 surface feet per minute. The metal removal rate shall be at least 7.5 cubic inches per minute.

3.10.3 Cylindrical turning. Using the same test bar as for the metal removal requirement (see 3.10.2), the specimen shall be semifinished to the configuration shown in Text 22 of ANSI B5.16. A finish cut shall be taken over the two collars in one pass using a single point carbide tip turning tool and a cutting speed of not less than 350 surface feet per minute. The turned diameter shall conform to the accuracy of Test 22 of ANSI B5.16. The same bar shall then be turned between

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centers
by taking a single cut taken over the two collars. The turned diameter shall conform to the accuracy of Test 23 of ANSI B5.16.

3.10.4 Taper turning. The taper attachment shall be used to cut a 10 degree taper over a length of at least 9 inches in a bar of low carbon steel (SAE 1020). The vertical deflecting of the cutting tool over the 9 inch taper length shall be no more than 0.002 inch while removing 0.062 inch of metal at a speed of 668 plus or minus 20 surface feet per minute.

3.10.5 Collet and drill chuck. The following diameters low carbon steel (SAE 1020) shall be gripped by an appropriate collet furnished with the lathe and the following drill bits shall be mounted in the appropriate drill chuck mounted in the tailstock holes of the following depths and shall be drilled using the tailstock handwheel.

Diameter of Bar Stock	Drill Size	Depth of Hole.
3/16 inch	#70	1/4 inch
3/4 inch	1/2 inch	1 inch

The work mounted in the lathe shall not slip and shall be held securely during drilling.

3.11 Equipment. The following equipment shall be provided with each lathe and shall all be capable of operating at maximum capacities of the lathe.

3.11.1 Thread chasing dials. The thread chasing dial, engaging the leadscrew, shall be attached to the carriage apron. The thread chasing dial shall be graduated to indicate the half nut engaging positions when threading inch threads.

3.11.2 Face plate, small. The small face plate shall be made of steel or steel alloy for mounting on the spindle nose, and shall have not less than two radial slots for driving a lathe dog. The slots shall be in accordance with ANSI B5.1. The face plate shall be not less than 9 inches in diameter. Face plate runout shall be in accordance with ANSI B5.16.

3.11.3 Face plate, large. The large face plate shall be made of steel or steel alloy, for mounting on the spindle nose. The face plate shall have no less than four T-slots or through slots in accordance with ANSI B5.1. The face plate shall be not less than 17 inches in diameter. Face plate runout shall be in accordance with ANSI B5.16.

3.11.4 Center sleeve. One headstock center sleeve shall be provided for adapting the taper in the spindle nose to the taper of the center supplied. Concentricity shall be within 0.0002 inches total indicator reading. Tapers shall be in accordance with ANSI B5.10 for self-holding tapers.

3.11.5 Lathe centers. Two lathe centers, one for the headstock and one for the tailstock, shall be furnished. All centers shall conform to ANSI B5.10 and table I requirements. The centers shall be hardened to Rockwell C-62 and C-68 and ground to within 0.0002 inch total indicator reading for concentricity.

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3.11.6 Indexing tool post. The tool post shall be square, indexing turret type for holding at least four tool holders. The tool post shall be surfaced hardened and ground to not less than Rockwell C-48. The tool pose shall have a 5/8 by 1 3/8 inch capacity size.

3.11.7 Chuck, 4-jaw, independent. The chuck body shall be made from a steel alloy and shall fit to the spindle nose without adapters other than as a part of the chuck. Four reversible step jaws with adjusting screws and wrench shall be included with each lathe. All working parts of the chuck shall be heat-treated. The total indicator reading on the periphery of the chuck, face of chuck body, and the face of the jaw steps shall be in accordance with the requirements of ANSI B5.16 and ANSI B5.8. The chuck shall be of medium duty and shall have an outside diameter of not less than 15 inches. A self-ejecting chuck wrench shall be provided for each chuck.

3.11.8 Chuck, 3-jaw, universal. The chuck shall be made from a steel alloy, shall be universal geared and self-centering. All working parts shall be heat-treated. Run out of the chuck periphery, face of the body, and jaw steps shall be in accordance with the requirements of ANSI B5.16 and ANSI B5.8. The chuck shall be of medium duty and shall fit the spindle nose without the use of adapters. The universal 3-jaw chuck shall have an outside diameter of not less than 12 inches. A self-ejecting chuck wrench shall be provided for each chuck.

3.11.9 Center, live. The live center shall be the ball or tapered roller bearing type for mounting in the tailstock spindle of the lathe. The live center shall be hardened and ground to not less than Rockwell C-50.

3.11.10 Dog, set, lathe. The set shall consist of five bent tail, safety screw lathe dogs. One lathe dog each of 1, 2, 3, 4, and 5 inch capacities shall be provided.

3.11.11 Steady rest. The steady rest shall fit the bed ways and shall have means for being clamped to the ways. The steady rest shall be of the hinged-type with three adjustable, bronze or roller tip jaws. The rollers shall run on antifriction, sealed bearings. All working parts shall be finished and shall contain adjusting and locking screws for each of the three jaws. The steady rest shall accommodate 1/2 through 2 1/2 inch diameter stock.

3.11.12 Follow rest. The follow rest shall be fitted to the carriage with means for quick installation and removal. The follow rest shall have two or more adjustable, bronze or roller tip jaws. The rest shall have adjusting screws and lock screws for setting and locking the jaws for any stock diameter from 1/2 to 2 1/2 inches.

3.11.13 Micrometer carriage stop. The micrometer stop shall be mounted on the front bed to provide adjustable stop. The micrometer dial shall be graduated in inch and metric increments of no more than 0.001 inches and 0.020 millimeters with at least 1 inch of adjustment.

3.11.14 Lathe tool holder sets. The lathe tool holders, as specified in 3.11.14.1 to 3.11.14.4, shall be fitted to the tool post (see 3.11.6). The shanks of the tool holders shall be hardened and ground to not less than Rockwell C-40. The specified tool holder sets shall include all tool bits, etc., necessary to establish proof that the tooling, as specified in 3.11.1 through 3.11.13, function and operate with the lathe.

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3.11.14.1 Lathe tool holder set, cut-off. The cut-off tool shall be a set of three tools consisting of one each: straight holder, right hand offset holder and a left hand offset holder. Each turning tool shall be complete with a high speed turning tool bit. The holder size shall be 5/8 by 1 3/8.

3.11.14.2 Lathe tool holder set, turning. The turning tool holder set shall be a set of three tools consisting of one each: straight holder, right hand offset holder, and left hand offset holder. Each turning tool shall be complete with a high speed turning tool bit. The holder size shall be 5/8 by 1 3/8.

3.11.14.3 Lathe tool holder set, boring bar. The boring bar tool holder set shall be complete with bars and holders. Two bars shall be supplied, one 1/2 inch bar for 3/16 inch square, high speed cutters and one 3/4 inch bar for 1/4 inch, square, high-speed cutters. One end of each bar shall be broached at a 90 degree angle with opposite end broached at a 45 degree angle. The holder size shall be 5/8 by 1 3/8.

3.11.14.4 Lathe tool holder, threading. The threading tool shall be a 60 degree sharp V-form cutter. The holder size shall be 5/8 by 1 3/8.

3.11.15 Special wrenches. One set of any special wrenches, required for operation and maintenance of the lathe, shall be provided.

3.11.16 Taper attachment. The taper attachment shall be a telescoping screw type for mounting in any working position on the rear of the lathe bed. It shall be graduated for setting in both degrees and inches of taper per foot. Turning capacity shall be not less than 15 inches in length in one setting and 16 degrees included angle. The attachment shall not interfere with the normal lathe operations when the taper attachment is not in use. The lathe attachment shall be rigid enough to limit tool deflection to not more than 0.0008 inch over a length of 9 inches when the taper attachment is being used to remove 0.062 inch thickness of material.

3.12 Information plates. Information plates shall be provided. All information plates specified herein (see 3.12.1 to 3.12.3) shall be written in the English language and shall be clearly and legibly etched, stamped, embossed, or engraved in bold face on a contrasting background on a corrosion resistant metallic composition plate and shall be securely attached to each lathe in a location visually convenient to the machine operator's normal work station. All plates shall be permanently secured to the lathe with fasteners such as screws, bolts, and rivets.

3.12.1 Nameplate. The nameplate and other information plate(s) can be incorporated into one or more plates at the manufacturer's option. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number
Contract number
Date of manufacture

3.12.2 Lubrication plate. The following information shall be included on the lubrication plate:

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Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal Specification number of lubricants

3.12.3 Instruction plates. Instruction plates shall be clear and concise in their meaning and application, with a special emphasis placed on removing any ambiguous terminology which may confuse or misdirect the machine operator as to the lather's functions, operations, or capacities. All instruction plates shall be located on the machine in a position that the lathe operator can readily and easily receive (read) the necessary instructions from the working station where the lathe operator is required to be positioned to engage the lathe or the lathe settings.

3.13 Workmanship. The quality of workmanship imparted to the lathe and all its components shall be equal to or exceed the quality of the current commercial lathe(s) manufactured by domestic producers of the type of lathe addressed herein. Where the requirements herein exceed commercial requirements, the requirements herein shall take precedence. The neatness, safety, sharp edges, structural integrity (seams, laps, steps, etc.), material soundness (no cracks, deformations, structural weakness, etc.), brazing and soldering, painting, welding, etc. of the lathe shall equal or exceed the requirements as stated herein. The quality of workmanship shall be such as to eliminate degradation of the form, fit, function, performance, and appearance as specified herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.2). Unless otherwise specified (see 6.2.1), first article inspections shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass any examinations and tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of a through i below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (4.5)
b. Center sleeve concentricity (4.6.1)
c. Lathe center concentricity (4.6.2)
d. Hardness tests (4.6.3)
e. Paint test (4.6.4)
f. Performance tests (4.6.5)
g. Alignment accuracy test (4.6.6)
h. Dielectric strength test (4.6.7)
i. Packaging inspection (4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105 using inspection levels and acceptable quality levels (AQL) as listed unless 100 percent inspection is specified. A sample lot shall consist of a quantity produced in 1 month or 150 lathes, whichever is less.

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	100 percent	1.0
Center sleeve concentricity (4.6.1)	II	1.0
Lathe center concentricity (4.6.2)	II	1.0
Hardness (4.6.3)	II	1.0
Paint test (4.6.4)	II	1.0
Performance tests (4.6.5)	II	1.0
Alignment accuracy test (4.6.6)	II	1.0
Dielectric strength test (4.6.7)	II	4.0
Packaging inspection (4.7)	S-4	4.0

4.5 Product examination. Visually, manually, and dimensionally examine the lathe to determine conformance with the requirements of 3.2 through 3.3.2, 3.3.4 through 3.7, and 3.13. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. Dimensional examination shall include verification that the lathe meets all specifications and characteristics listed in table I. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the lathe has passed the visual, manual, and dimensional verification prescribed for it by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.3.1), shall be cause for rejection.

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4.6 Tests. Tests shall be performed as specified in 4.6.1 through 4.6.7. Measuring instruments used in performing the following tests shall be more accurate by a factor of ten than the specified limit, and shall be traceable to the National Bureau of Standards.

4.6.1 Center sleeve concentricity. The center sleeve shall be inserted in the headstock and shall be checked for concentricity. Concentricity shall be within 0.0002 inch total indicator reading and failure of the sleeve to meet requirement shall be cause for rejection (see 3.11.4).

4.6.2 Lathe center concentricity. The two lathe centers, one for the headstock and one for the tailstock, shall be checked for concentricity. Concentricity shall be within 0.0002 inch total indicator reading and failure to meet requirement shall be cause for rejection (see 3.11.5).

4.6.3 Hardness. Rockwell C hardness tests shall be performed on the components of the lathe listed below, or on test samples processed currently with the components. Failure of the component or sample to meet the hardness number shown for that component shall be cause for rejection. A Certificate of Conformance (COC) may be accepted for a component in lieu of the test, provided the COC is written in the English language and is signed by a responsible officer of the original equipment manufacturer of that component.

Component	Rockwell C No. (min)	Ref Para.
Gears (all)	48	3.2.3
Bed ways	42	3.5.1
Spindle nose, headstock	50	3.5.2.1
Lead screw	38	3.5.3
Cross feed screws	45	3.5.6
Spindle, tailstock	48	3.5.8
Centers, lathe	62-68	3.11.5
Tool post	48	3.11.6
Center, live	50	3.11.9
Tool holders (all)	40	3.11.14

4.6.4 Protective finish. The hardness, adhesion, and thickness of the paint used on the lathe shall be checked as follows:

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the lathe until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

b. A test sample of the same metal, painted with the same process and at the same time as the lathe, shall have a sharpened H lead pencil pressed into the paint as in a. Failure of the surface to pit (paint too hard) shall be cause for rejection.

c. Two (2) inch intersecting lines shall be cut through the paint

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to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection (see 3.3.3).

4.6.5 Performance. The lathe shall be subjected to the following turning tests after the machine has reached operating condition. Failure of any test shall be cause for rejection (see 3.10).

4.6.5.1 Turning. A bar of low carbon steel SAE 1020 not less than 2 inches in diameter and 4 inches long shall be in the 3-jaw chuck supplied with the lathe. A rough and finish cut shall be made at least 2 inches long using a cutting speed of not less than 300 surface feet per minute. The turned diameter shall conform to the accuracy of Test 21 in ANSI B5.16 (see 3.10).

4.6.5.2 Metal removal turning. A bar of low carbon steel SAE 1020. The vertical deflection of the cutting tool over the 9 inch taper length shall be no more than 0.002 inch while removing 0.062 inch of metal at a speed of 668 plus or minus 20 surface feet per minute (see 3.10).

4.6.5.5 Collet and drill chuck. A length of 3/16 inch diameter low carbon steel SAE 1020 shall be gripped by an appropriate collet. A #70 drill bit shall be inserted in the drill chuck and the chuck inserted in the tailstock. Using the tailstock hand wheel, drill a #70 hole 1/4 inch deep. Remove the collet and drill bit. Install another collet furnished with the lathe and grip a length of 3/4 inch round low carbon steel SAE 1020. Insert 1/2 inch drill bit in the drill chuck and drill a hole 1 inch deep in the steel. Using the hand wheel, back the drill out of the work until the drill chuck is ejected. Failure of the collets to prevent slippage of the work pieces, failure of the drill chuck to hold the bit securely, or failure of the chuck to eject when the tailstock was retracted shall be cause for rejection (see 3.10).

4.6.5.6 Cutting power. Using a round medium carbon steel SAE 1045 bar of not less than 2.00 inches in diameter and having a minimum hardness of Brinell 160, perform cutting tests as listed.

4.6.5.6.1 Rough cutting. The rough cut shall be no less than .080 inch deep and no less than 3.00 inches long. The rough cut shall be made at a feed rate of not less than 0.015 inch per revolution at a cutting speed of not less than 350 surface per minute (see 3.9.1). The rough turned diameter shall show no evidence of chatter and shall meet a total tolerance requirement of .005 inch or less for both out-of-round and taper per foot. Failure to meet requirements shall be cause for rejection.

4.6.5.6.2 Finish cutting. Finish cutting operation shall be performed on the same piece of steel after the rough cutting test. The finish cut shall be no less than .020 inch deep and no less than 3.00 inches long. The finish cut shall be made at a feed rate of not less than .005 inch per revolution at a cutting speed of not less than 400 surface feet per minute (see 3.9.2). The finished diameter shall meet the requirements specified in ANSI B5.16 (see 3.9.2).

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4.6.6 Alignment accuracy. The lathe shall be tested to ensure conformance to accuracy and alignment tolerances as specified in ANSI B5.16 Tests 1 through 20 both before and after being subjected to tests specified in paragraphs 4.6.5.6.1 and 4.6.5.6.2 (see 3.8).

4.6.7 High voltage. The entire electrical system shall be subjected to 2500 volts rms at 60Hz applied instantaneously between mutually insulated circuits, between insulated circuits and ground, and between conductor and the chassis ground. The 2500 volts rms shall be held for at least 15 seconds using at least a 500 volt ampere transformer. During the 15 seconds at test voltage the circuitry shall be observed for flashover (surface discharge), sparkover (air discharge), and breakdown (puncture discharge). Any evidence of flashover, sparkover, breakdown, or damage to the unit due to heat or other effects of the applied voltage, shall be cause for rejection (see 3.6.4).

4.6.8 Electromagnetic interference control. When specified (see 6.2.1 and 6.6), the lathe shall be tested for compliance with MIL-STD-461 in accordance with the tests and procedures of MIL-STD-462.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the preservation, packing, and marking requirements specified in section 5.

5 PACKAGING

5.1 First article. When specified (see 4.3 and 6.2.1), the contractor shall furnish one complete pack for first article packaging inspection. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements stated herein. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements stated herein.

5.2 Disassembly. When practical, items shall be disassembled to afford protection of components, attachments, and accessories against damage and pilferage and to reduce cubage. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identify proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Matchmarking. Disassembled parts shall be matchmarked when necessary to facilitate reassembly. Removed parts and mating parts on the item shall be matchmarked by stenciling on the part or by the use of tags. Tags shall be waterproof and ink used shall be waterproof. Tags shall be firmly secured to each individual part.

5.4 General requirements. Component parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered herein shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures, and preservatives shall conform to the criteria and guidelines specified herein.

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5.5 Level A preservation. When specified (see 6.2.1) in the contract, preservation shall be level A.

5.5.1 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces shall be performed before the application of a preservative. Care shall be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts.

5.5.2 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9 of VV-L-800. To prevent dust from accumulating on cleaned parts, the parts shall be covered with a sheet of chemically neutral paper.

5.5.3 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, etc., shall be removed from the assembly prior to the external cleaning operation. All external, critical ferrous metal surfaces, fit, or function, of the machine shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. Noncritical ferrous metal surfaces shall be solvent cleaned and wiped or drain dried. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic composites. When cleaned surfaces of the machine show signs of rusting before application of the final preservative, reprocessing surface shall be confined to that area of the assembly. Alternatively, the solvent used for the final wiping operation shall contain a small amount, approximately 5 percent by volume, preservative oil conforming to P-10, type 10, grade 30, of MIL-L-21260. It is extremely important that elapsed time between these operations be held to an absolute minimum.

5.5.4 Steam and air lines. Steam and air lines shall be drained of all water accumulation and dried by injecting moisture-free air into the systems.

5.5.5 Disassembly components. Disassembled components detached from the basic machine shall be cleaned and dried as specified herein for like items.

5.5.6 Machine, metal and woodworking tools, and tool accessories. All such tools and tool accessories shall be clean and free from contaminants. Preservatives shall be applied to provide protection from corrosion during shipment and storage. All preserved items shall be wrapped with greaseproof barrier material. Critical surface items, i.e., parallels, etc., shall also be wrapped with no less than 1/2 inch uncompressed cushioning material.

5.5.7 Electrical equipment. Electrical equipment shall be clean and dry. Preservatives shall be applied as required to prevent corrosion to moving or other ferrous metal parts, but shall not come into contact with electrical connections, windings, etc. Openings shall be sealed with cap-plugs or with waterproof, pressure sensitive tape to provide protection against entry to any foreign materials.

5.5.8 Hoses and fittings. Open ends of all hose or hose fittings shall be sealed with plastic caps or plugs or with waterproof, pressure sensitive tape that will provide protection against entry of any foreign material. Attached hoses shall be coiled and secured to prevent hose from unwinding. Hoses not attached to the

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machine shall be coiled uniform, compact and of a diameter to prevent deformation or kinking. Each coiled hose shall be secured approximately equidistant a minimum of three places with cotton tape, twine or pressure sensitive tape. Loose fittings shall be placed in a transparent plastic bag, sealed and secured on the inside of the coil.

5.5.9 Lubricating systems. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with each machine. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT." The operator's manual shall also reflect the above warning and procedure.

5.5.10 Maintenance and hand tools. Maintenance and hand tools such as screwdrivers, hammers, chisels, etc., shall be clean and free from contaminants. Sharp edges such as cutting edges shall be protected by a fiberboard shield placed on the edge and taped in place. Tools shall be placed in a fiberboard box and secured from movement so as to prevent damage to the tools or other items.

5.5.11 High-speed spindle bearings. Fill the reservoir to normal operating level with P-9, VV-L-800, preservative oil. If the system is of such design that the oil will leak out during shipment or storage, the system shall be preserved by filling with preservative oil conforming to P-10, type I, grade 30, of MIL-L-21260 and drained.

5.5.12 Hydraulic systems. Both open and closed-type systems shall be filled to normal operating levels with manufacturer's hydraulic fluid oil. Filling of the systems shall be accomplished while the machine is connected to the power source. After filling, the machine shall be cycled in order to circulate the oil through the system. Prior to disconnecting the machine from the power source, place the piston(s), except opposing pistons, in their innermost position. The oil shall remain in the reservoir, if possible, and sealed.

5.5.13 Grease fittings. Charge all grease fittings with preservative compound conforming to P-11 of MIL-G-10924.

5.5.14 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with grade 2 preservative oil conforming to P-2, of MIL-C-16173, grade 2 preservative.

5.5.15 Gear cases. Fill the gear cases with normal lubricating oil. Shift gears into all possible positions while operating the machine, then close and seal all openings.

5.5.16 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2 of MIL-C-16173, grade 2 preservative compound and covered with greaseproof barrier material. Parts of the machine which are adjustable shall be locked into place.

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5.5.17 Bearings and journals. Grease lubricated bearings shall be preserved with P-11, MIL-G-10924 preservative. Oil lubricated journals shall be filled with lubricating oil conforming to P-10, type I, grade optional, of MIL-L-21260.

5.5.18 Drive belts and pulleys. Drive belts shall either be removed from the machine or released from tension. Ferrous metal pulleys shall be coated with rust-inhibiting primer. Belts removed from the machine shall be packaged in a waterproof, greaseproof bag and sealed.

5.5.19 Electrical wiring system. Exposed ends of wires, plug openings, sockets, connector plugs, terminals, and openings into switches and junction boxes, etc., shall be sealed with waterproof tape or cap-plugs, respectively. Electric cable shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling.

5.5.20 Closed dial indicators. Closed dial indicators shall be covered with 1 inch thick uncompressed cushioning material and secured in place with waterproof, pressure sensitive tape. Cushioning materials and tape sued shall be water-resistant, nondusting and noncorrosive.

5.5.21 Drawers and door assemblies. Sliding surfaces of drawer guides shall be coated with P-11 of MIL-G-10924, preservative grease. Interior surfaces of door hinges shall be lubricated by application of P-10, grade 30, of MIL-L-21260, preservative oil. In addition to mechanical locks or catches incorporated in the machine tool, secure doors and drawers with metal strapping or other devices which are suitable for the purpose intended. Fiberboard pads shall be placed under metal strapping to prevent marring the surfaces.

5.5.22 Closure of openings. Small openings, except vents and louvers installed for ventilation purposes, shall be sealed with waterproof, pressure sensitive tape. Large openings shall be covered with waterproof barrier material. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render coverings vulnerable to puncture, coverings shall be protected by applying wood, plywood, sheet metal or fiberboard over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, waterproof tape or plastic cap-plugs.

5.5.23 Technical data. Technical data furnished with the machine shall be packaged in a waterproof, greaseproof barrier material and sealed. The package shall be legibly and clearly marked "TECHNICAL DATA" in bold letters. The package shall be protected against puncture and abrasion. The data shall be either attached to the machine with tape, placed in a stowage compartment of the machine, or positioned in a consolidated box containing accessories of disassembled components.

5.5.24 Consolidated packaging. Packaged accessories, maintenance tools, and small disassembled components shall be consolidated and placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in fiberboard box or boxes, class weather-resistant, of sufficient strength to handle weight inside. Large bulky disassembled components shall be packed as specified (see 5.6.1).

5.5.25 Cushioning. Cushioning materials shall be water resistant, nondusting and noncorrosive, and of sufficient thickness to prevent damage, not less than 1/2 inch thick wrap.

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5.5.26 Protection of preservatives. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with greaseproof barrier material. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. Secure barrier in place with waterproof, pressure sensitive tape.

5.5.27 Movable parts. Operating heads, sliding tables, rams, counter-balances, and other movable parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and , therefore, must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels, cranks, levers, and similar controls shall be secured in position with common annealed wire or steel strapping.

5.5.28 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered, blocked and braced to prevent movement.

5.5.29 Scales. Measurement, indicating, etc., scales shall be individually enclosed with no less than 1 inch uncompressed cushioning and places in a snug-fitting fiberboard box. To provide maximum protection, the fiberboard box(es) shall be placed in a wooden box conforming to PPP-B-601, domestic, style optional, for later consolidated into shipping container.

5.6 Level A packing. When specified (see 6.2.1) in the contract, packing shall be level A.

5.6.1 Machinery weighing 2,500 to 30,000 pounds. Machines weighing 2,500 to 30,000 pounds shall be packed individually in crates conforming to MIL-C-104, type I, class 2, style A. Consolidated containers shall be packed with the machine in the specified crate. Arrangement of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with the appendix to the crate specification, MIL-STD-1186, and as specified herein. Consolidated containers for large bulky disassembled items (see 5.5.25) shall conform to PPP-B-601, domestic, style optional.

5.6.2 Leg-type machines. Leg-type machines with slotting holes for mounting shall be reinforced by placing a nominal 2 by 4 inch wood block against the leg opposite the slot opening and nailing to the skid base.

5.6.3 Bolting. Machines shall be secured to the skid base by not less than four bolts. When sufficient bolting points are not available, J-clamps or tie rod assemblies, or other devices, shall be used to provide the minimum of four holding points. Bolts shall pass completely through the skid runner or sleeper, as applicable, and the base flooring.

5.6.4 Consolidated containers. Consolidated containers shall be placed inside the shipping container housing the machine. The boxes shall be restrained by use of metal strapping or tie rod assemblies.

5.6.5 Operational oils. Operational oils furnished with the machine, but not placed in the machine stowage receptacles, shall be stowed in metal drum(s). Drums shall be positioned upright inside the shipping container housing the machine. Blocking shall be applied to immobilize the drum(s) inside the shipping container.

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5.7 Military marking. When specified (see 6.2.1) in the contract, marking shall be in accordance with MIL-STD-129. Bar coding is also required in accordance with MIL-STD-129.

5.8 Level C preservation, packing, and marking. When specified (see 6.2.1) in the contract, preservation, packing, and marking shall be in accordance with the requirements of MIL-STD-1190. Bar coding shall be applied. In addition to the requirements of MIL-STD-1190, all unprotected ferrous metal parts and all accessories shall e cleaned, preserved and wrapped with greaseproof wrap. Accessories shall be packed, blocked, and cushioned in fiberboard container(s). Any critical surfaces shall be protected from damage in transit. Technical data shall be placed in a plastic bag (stapled or sealed closed) and enclosed in the accessory container. Movable parts of the machine shall be secured by mechanical devices incorporated into the machine for the purpose. They must also be augmented with appropriate wood blocking, metal strapping, etc.., to assure immobilization. Levers, cranks, handles, etc., shall be secured in position with annealed wire or steel strapping and protected from damage. All items shall be protected from marring.

5.9 Packing for lubrication systems. The lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that all temporary caps or plugs must be removed before start-up or use of the lathe, shall be attached to all temporary caps or plugs. These warnings shall also be included in the instruction manual.

5.10 Workmanship. Workmanship shall be such that when proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and suitable for long term storage.

6 NOTES

6.1 Intended use. The lathe covered by this specification is intended for use in machine shops for fine and rough metal turning and boring, and for cutting screw threads, fine and course, right-hand or left-hand, as specified herein.

6.2 Order data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. First article, when required (see 3.1).
c. First article inspection, when required (see 4.3 and 5.1).
d. Packaging required (see 5.5, 5.6, 5.7, and 5.8).

6.3 First article. When a first article inspection is required, the lathe(s) should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those

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bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approved is presently appropriate for the pending contract.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.5 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification should also include the name of the original equipment/part manufacturers, Contractor and Government Entity (CAGE), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.6 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation and other health and safety requirements, should be specified by the user.

6.7 Subject term (key word) listing.

Brazing
Castings
Dielectric Strength
Fastening Devices
Finish
Forging
Gears
Hardness
Interchangeability
Lathes
Safety
Soldering
Threads
Welding
Workmanship

6.8 Change from previous issue. Asterisks are not used in the revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL

Review activities:
DLA -

User Activities:
Army -

Preparing activity:
Army - AL

Project No. 3416-A208

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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